Operational Research II, Final Exam, 2026 Spring

Solution and Grading

- Duration: 120 minutes
- Closed material, No calculator

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- Write legibly.
- Justification is necessary unless stated otherwise.

1	20
2	20
3	30
Total	70

#1. Assume that calls arrive at a customer call center between 9am ($t = 0$) and 2pm ($t = 5$) following a non-homogeneous Poisson process with the rate function of the following:

$$\lambda(t) = \begin{cases} 4 & 0 \le t < 1 \\ 2t + 2 & 1 \le t < 3 \\ 8 & 3 \le t < 5 \end{cases}$$

(a) What is the probability that the call center receives 7^{th} call before 10am ($t = 1$)? `[10pts]`

(b) What is the expected number of arrivals between 11am ($t = 2$) and 1pm ($t = 4$)? `[10pts]`

Solution:

- (a) Let X be the number of received calls.

$$\begin{aligned}
\mathbb{P}\left(N(1) - N(0) \ge 7\right) &= 1 - \mathbb{P}(N(1) - N(0) < 7) \\
&= 1 - \mathbb{P}(X < 7 | X \sim Poi(\int_0^1 4\ dt)) \\
&= 1 - \mathbb{P}\left(X < 7 | X \sim Poi(4)\right) \\
&= 1 - \left(\mathbb{P}\left(Poi(4) = 0\right) + \mathbb{P}\left(Poi(4) = 1\right) + \mathbb{P}\left(Poi(4) = 2\right) + \cdots + \mathbb{P}\left(Poi(4) = 6\right)\right) \\
&= 1 - \left(\frac{4^0 e^{-4}}{0!} + \frac{4^1 e^{-4}}{1!} + \frac{4^2 e^{-4}}{2!} + \cdots \frac{4^6 e^{-4}}{6!}\right) \\
&= 1 - (1 + 4 + 8 + (32/3) + (32/3) + (128/15) + (256/45)) \cdot e^{-4} \\
&= 1 - \frac{437}{9} e^{-4}
\end{aligned}$$

- (b) Let, X is the number of arrivals. Then X follows Poisson distribution with parameter $\int_2^4 \lambda(t)\ dt$.

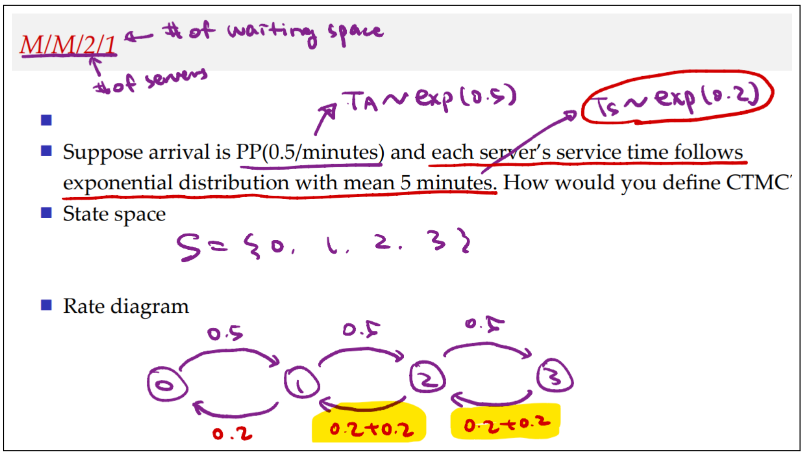
$$\begin{aligned}
\int_2^4 \lambda(t)\ dt &= \int_2^3 2t + 2\ dt + \int_3^4 8\ dt \\
&= \left[t^2 + 2t\right]_2^3 + 8(4 - 3) \\
&= (15 - 8) + 8 \\
&= 15
\end{aligned}$$

Since $X \sim Poi(15), \mathbb{E}X = 15$.

Grading scheme:

- (a) 5pts per mistake is deducted only if the mistake is considered a minor mistake.
- (b) 5pts per mistake is deducted only if the mistake is considered a minor mistake.

#2. Consider the following lecture note content. To define the $M/M/2/1$ queue as a CTMC, the state is defined as the number of customers in the system, yielding the state space $S = \{0, 1, 2, 3\}$. A corresponding rate diagram has also been constructed.



(a) In the rate diagram, the highlighted portions are labeled as "0.2 + 0.2". This indicates that when there are 2 or 3 customers in the system, both servers are busy, and thus the rate at which the number of customers decreases is the sum of the service rates of the two servers. Formally state the relevant theorem that justifies this modeling approach. (Hint: The theorem is related to the exponential distribution.) `[10pts]`

(b) Prove the theorem you stated above. `[10pts]`

(a) If $X_1 \sim \exp(\lambda_1)$ and $X_2 \sim \exp(\lambda_2)$ and they are independent, then $\min(X_1, X_2) \sim \exp(\lambda_1 + \lambda_2)$.

Grading criteria:

- The **random variables** (e.g., X_1, X_2) and their **distribution parameters** (e.g., λ_1, λ_2) must be correctly stated and clearly distinguished. If failed to do so, then 0pt.
- The theorem must be written in a clear **"if–then"** structure. If failed to do so, then -5pts.
- If the assumption of "independence" is missing, then -2.5pts.

(b)

$$\begin{aligned}
\mathbb{P}\big(\min(X_1, X_2) \le x\big) &= 1 - \mathbb{P}\big(\min(X_1, X_2) > x\big) \\
&= 1 - \mathbb{P}(X_1 > x,\ X_2 > x) \\
&= 1 - \mathbb{P}(X_1 > x)\,\mathbb{P}(X_2 > x) \quad \text{(by independence of } X_1, X_2\text{)} \\
&= 1 - \big(1 - \mathbb{P}(X_1 \le x)\big)\big(1 - \mathbb{P}(X_2 \le x)\big) \\
&= 1 - \big(1 - (1 - e^{-\lambda_1 x})\big)\big(1 - (1 - e^{-\lambda_2 x})\big) \\
&= 1 - \big(e^{-\lambda_1 x}\big)\big(e^{-\lambda_2 x}\big) \\
&= 1 - e^{-\lambda_1 x} e^{-\lambda_2 x} \\
&= 1 - e^{-(\lambda_1 + \lambda_2)x}
\end{aligned}$$

Thus, $\min(X_1, X_2) \sim \exp(\lambda_1 + \lambda_2)$.

Grading Criteria

- No partial credit.
- The proof must be based on the **cumulative distribution function (CDF)**. Proofs using other arguments (e.g., expectation) will not receive any credit.

#3. Consider a very small McDonald' s drive-through. An arriving vehicle must go through two stations: an **order** station, where the vehicle places an order and makes payment, and a **pickup** station, where the vehicle receives its food.

The movement of vehicles is described as follows.

- If the **order** station is empty upon arrival, the vehicle immediately enters the **order** station and begins service.
- If an arriving vehicle finds the **order** station occupied, it leaves the facility immediately. In other words, there is no waiting space.
- When a vehicle completes service at the **order** station and finds the **pickup** station empty, it proceeds to the **pickup** station.
- When a vehicle completes service at the **order** station and finds the **pickup** station occupied, the vehicle remains at the **order** station and waits until the **pickup** station becomes empty. In this case, the **order** station is still occupied, so no newly arriving vehicle cannot enter the **order** station.
- When a vehicle reaches the **pickup** station, it receives its food and then immediately leaves the facility as soon as completing service.

The following assumptions are made for this queueing system.

- Each station can serve at most one vehicle at a time.
- Vehicles arrive according to a Poisson process with rate 1 vehicle per minute.
- The **order** station can process, on average, 1 vehicle per minute, and the **pickup** station can also process, on average, 1 vehicle per minute. The service times at both stations are exponentially distributed.
- All interarrival and service times are independent.

(a) Define and describe the states of the CTMC. Then, draw the transition diagram[1]. `[10pts]`

(b) In the long run, what percentage of arriving customer vehicles receive the service? `[10pts]`

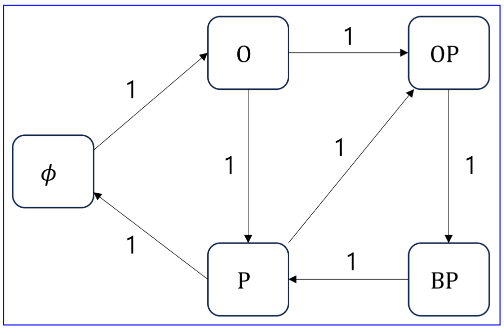
(c) The owner is considering adding one waiting space before the **order** station. If the same type of CTMC analysis is to be performed for this modified system, define the additional states that must be introduced. `[10pts]`

[1]Hint: Your state definition should distinguish between the following two cases: i) a vehicle at the order station is currently placing an order; ii) a vehicle at the order station has already finished ordering but is blocked because the pickup station window is occupied.

(a) The CTMC state space is $\mathcal{S} = \{\phi, O, P, OP, BP\}$. Writing out the description as below is necessary.

- ϕ: zero vehicle in the facility.
- O: 1 vehicle at the order station.
- P: 1 vehicle at the pickup station.
- OP: 1 vehicle at the order station, in the middle of receiving service . 1 vehicle at the pickup station.
- BP: 1 vehicle at the order station, completed ordering but is blocked and waiting for pickup station to clear. 1 vehicle at the pickup station. is at the pickup station.

The transition diagram is as follows:



(b) The rate matrix Q can be written as

$$Q = \begin{array}{c|ccccc} & \phi & O & P & OP & BP \\ \hline \phi & -1 & 1 & & & \\ O & & -1 & 1 & & \\ P & 1 & & -2 & 1 & \\ OP & & 1 & & -2 & 1 \\ BP & & & 1 & & -1 \end{array}$$

The stationary distribution π satisfies $\pi Q = 0$ and $\sum_{i \in \mathcal{S}} \pi_i = 1$. The flow balance equations are

$$\begin{aligned} -\pi_\phi + \pi_P &= 0, \\ \pi_\phi - \pi_O + \pi_{OP} &= 0, \\ \pi_O - 2\pi_P + \pi_{BP} &= 0, \\ \pi_P - 2\pi_{OP} &= 0, \\ \pi_{OP} - \pi_{BP} &= 0. \end{aligned}$$

From these equations, $\pi_\phi = 2\pi_{OP}, \pi_P = 2\pi_{OP}, \pi_{BP} = \pi_{OP}, \pi_O = 3\pi_{OP}$, and $\pi_\phi = \frac{2}{9}, \pi_O = \frac{3}{9}, \pi_P = \frac{2}{9}, \pi_{OP} = \frac{1}{9}, \pi_{BP} = \frac{1}{9}$. A newly arriving customer will be accepted to the system only if the order station is empty. Thus, 44.4% ($=\pi_\phi + \pi_P$) of arriving customers are accepted in the long run.

(c) In order for the waiting space to be occupied, the order station must be occupied as well. Thus, the additional states of "WO", "WOP", and "WBP" are necessary, in which the "W" obviously stands for the situation that waiting space is occupied by one vehicle.

Grading Criteria

- (a) If failed to distinguish "OP" and "BP", but the presentation with four states are well done, then 5pts is given.
- (b) If the state definition in (a) is not with the correct five states, then 0 pts for this problem.
- (c) If additional states are well described in accordance with your state definition in (a), then 10pts are given. Partial credit of 5pts may be given for genuine efforts.

[다음은 3번 문제의 한글 번역이다]

매우 작은 맥도날드 드라이브스루를 고려하자. 도착한 차량은 두 개의 station을 순서대로 거쳐야 한다. 첫 번째는 차량이 주문과 결제를 수행하는 **order** station이고, 두 번째는 차량이 음식을 수령하는 **pickup** station이다. 차량의 이동 방식은 다음과 같다.

1. 도착한 차량이 **order** station이 비어 있는 것을 발견하면, 그 차량은 즉시 **order** station으로 진입하여 서비스를 시작한다.
2. 도착한 차량이 **order** station이 점유되어 있는 것을 발견하면, 그 차량은 즉시 facility를 떠난다. 즉, waiting space는 존재하지 않는다.
3. **order** station에서 서비스를 완료한 차량이 **pickup** station이 비어 있는 것을 발견하면, 그 차량은 **pickup** station으로 이동한다.
4. **order** station에서 서비스를 완료한 차량이 **pickup** station이 점유되어 있는 것을 발견하면, 그 차량은 **pickup** station이 빌 때까지 **order** station에 남아 기다린다. 이 경우 **order** station은 여전히 점유된 상태이므로, 새로 도착한 차량은 **order** station에 진입할 수 없다.
5. 차량이 **pickup** station에 도달하면 음식을 수령하고, 서비스를 완료하는 즉시 facility를 떠난다.

이 대기행렬 시스템에 대해 다음과 같은 가정을 둔다.

- 각 station은 한 번에 최대 한 대의 차량만 서비스할 수 있다.
- 차량은 분당 1대의 비율을 갖는 Poisson process에 따라 도착한다.
- **order** station은 평균적으로 분당 1대의 차량을 처리할 수 있으며, **pickup** station도 평균적으로 분당 1대의 차량을 처리할 수 있다. 두 station에서의 서비스 시간은 모두 exponential distribution을 따른다.
- 모든 interarrival time과 service time은 서로 독립이다.

(a) CTMC의 state를 정의하고 설명하라. 그런 다음 transition diagram을 그려라[2]. `[10pts]`

(b) 장기적으로, 도착한 customer 차량 중 몇 퍼센트가 full service를 받는가? `[10pts]`

(c) 점주는 **order** station 앞에 waiting space를 하나 추가하는 방안을 고려하고 있다. 이 수정된 시스템에 대해 동일한 방식으로 CTMC 분석을 수행하고자 할 때, 새롭게 추가되어야 하는 state들을 정의하라. `[10pts]`

[2]힌트: State를 정의할 때 다음 두 경우를 구분해야 한다. i) **order** station에 있는 차량이 현재 주문을 수행하고 있는 경우; ii) **order** station에 있는 차량이 이미 주문을 완료했지만, **pickup** station이 점유되어 있어 blocked 상태에 있는 경우.